



Remote Team Member's Productivity Trends for Year 2025

Capstone - LGCC Data Analytics Cohort 6 (2025-2026)

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Methodology: Use of Python to Analyze Dataset of 2025

Data Source:

Internal Confidential
dataset

File Type: CSV

Time period: 2025

Data- Frame Shape:
Rows: 99203
Columns: 15

Python Libraries

Import pandas as pd



Import numpy as np



Import seaborn as sns



Import matplotlib.pyplot as plt

Definitions

- ▶ **Clinical review:** A medical professional reviews a proposed treatment or medication against insurance policy's coverage guidelines before care is provided. The goal is to ensure the treatment is medically necessary and covered.
- ▶ **Determination:** The insurance company either **approves** or **denies** those request.
- ▶ **Notification:** A formal communication (which includes verbal & Written) from health insurance company to member and provider confirming that they have received, reviewed, and have approved/denied a request from your doctor for a specific medication, procedure, or medical service. With verbal notification we must make 2nd attempt, if 1st attempt is unsuccessful.

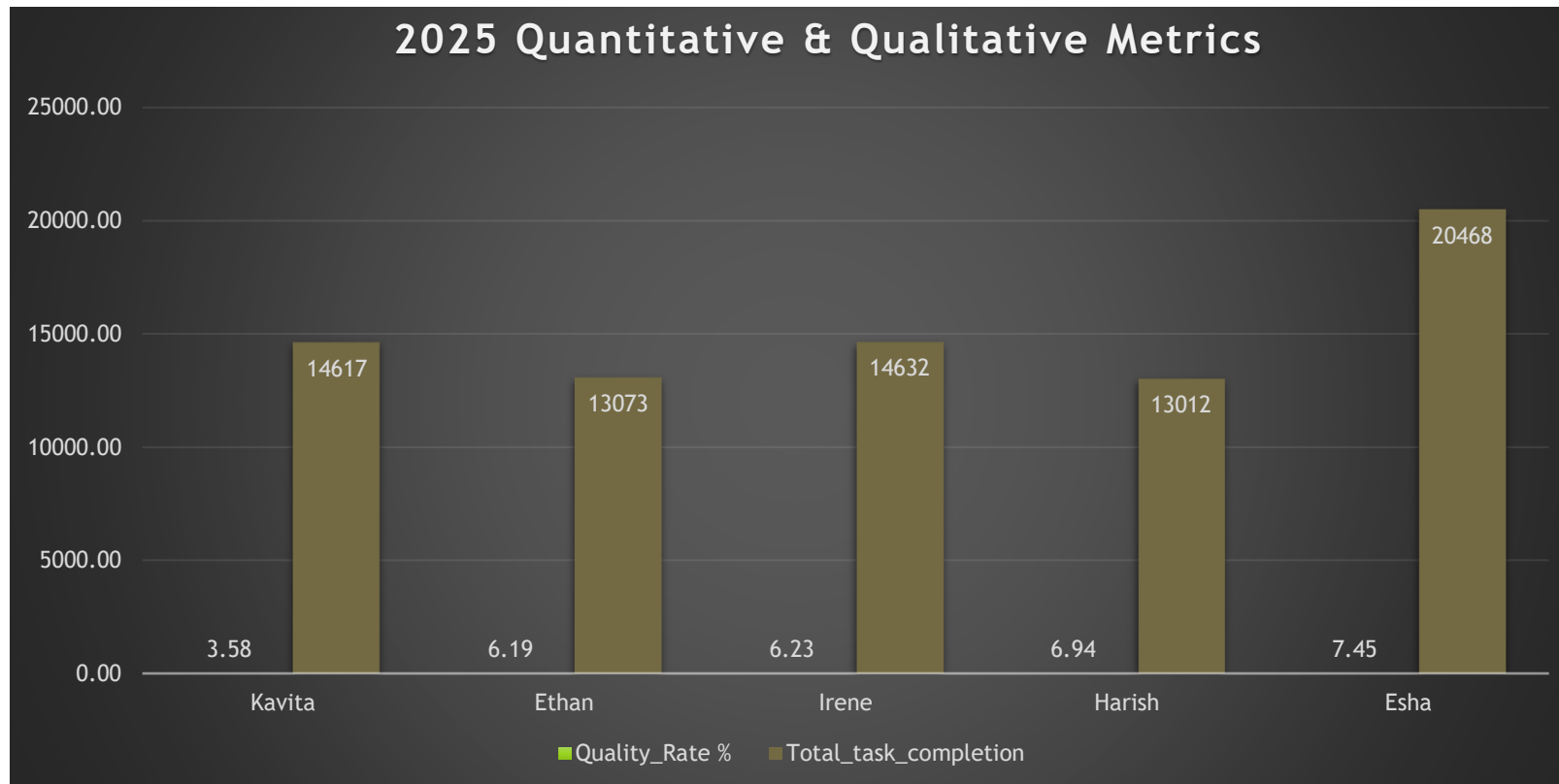
*** Turn around time for these requests are 24 to 72 hours****

2025 Quantitative & Qualitative Metrics

- ❖ Compare task completion volume & quality trends between team members

Tenure

- ❖ Top performers - employee for 0-2 years
- ❖ Under/average performers - employee for 5+ years



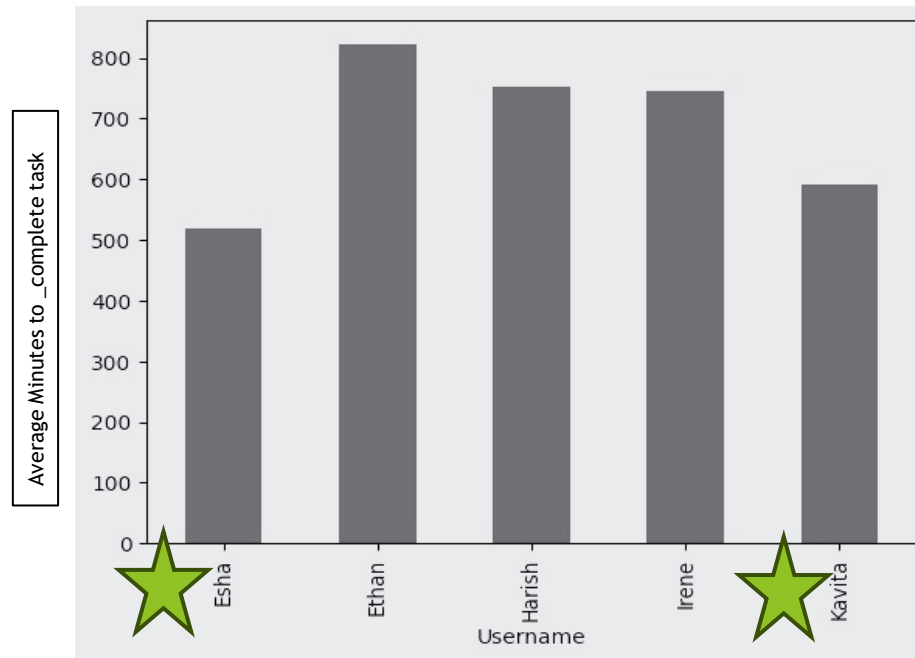
Objectives

The Purpose of this project is to analyze the data to identify the productivity trends within the remote team, specifically:

- Identify average time utilized to complete each task
- Average time utilized to complete core tasks
- Average total # of tasks completed by team members on daily as well as monthly basis
- Determine factors influencing higher number of tasks not completed
- Identify areas of opportunity to increase productivity

Overall Average Time Utilized to Complete a Task

Username	Average # of Minutes to Complete a Task
Esha	518
Ethan	822
Harish	751
Irene	746
Kavita	591



- Average time to complete a task was 685 mins
- Esha & Kavita task completion was within 75 to 85 % of the average time
- Irene, Harish & Ethan task completion was within 110 to 120 % of average time

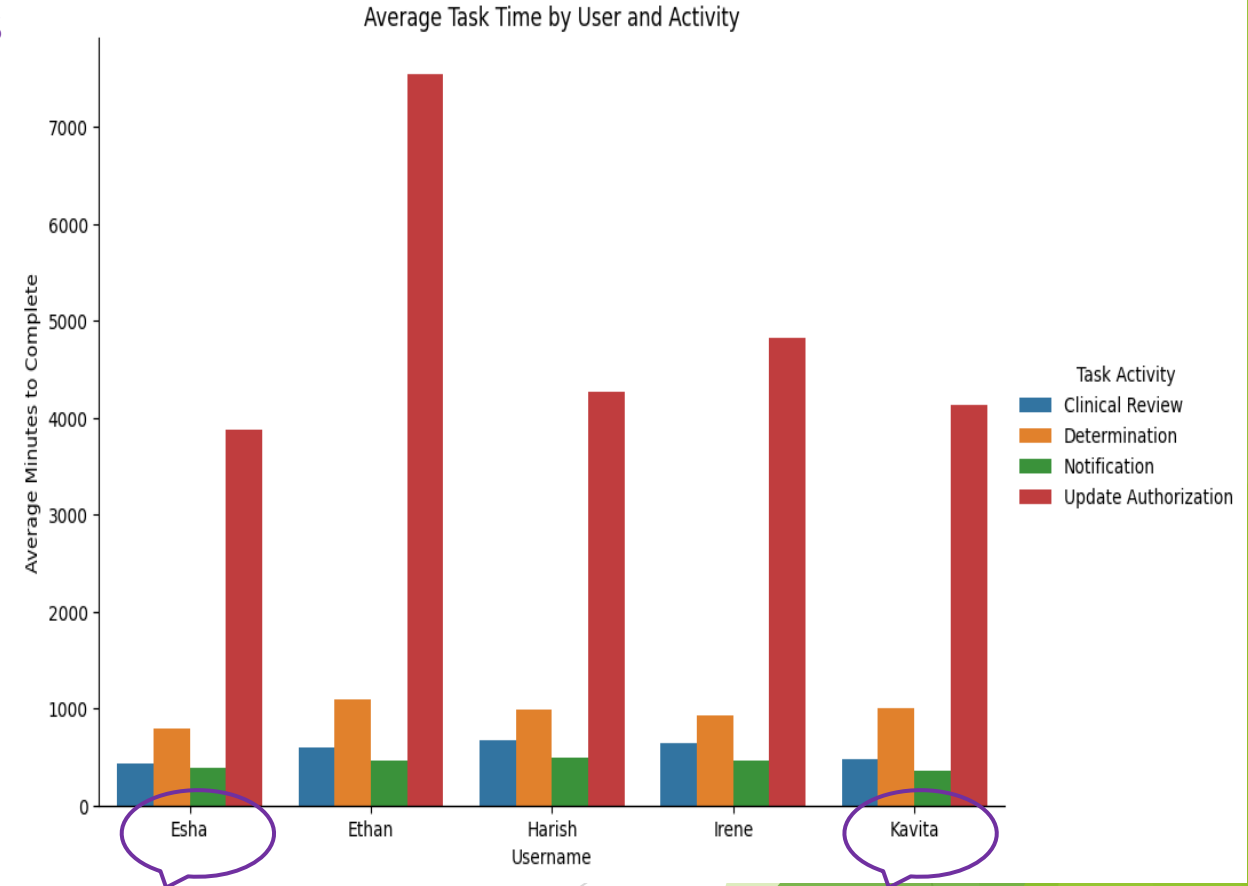
Average Time Utilized to Complete Core Tasks

➤ Average time to complete Notification Task was 462 mins

- Esha & Kavita - 75 to 85 % of average time
- Ethan & Irene - little over but close to average time
- Harish - 110 % of average time

➤ Average time to complete Clinical Review Task was 577 mins

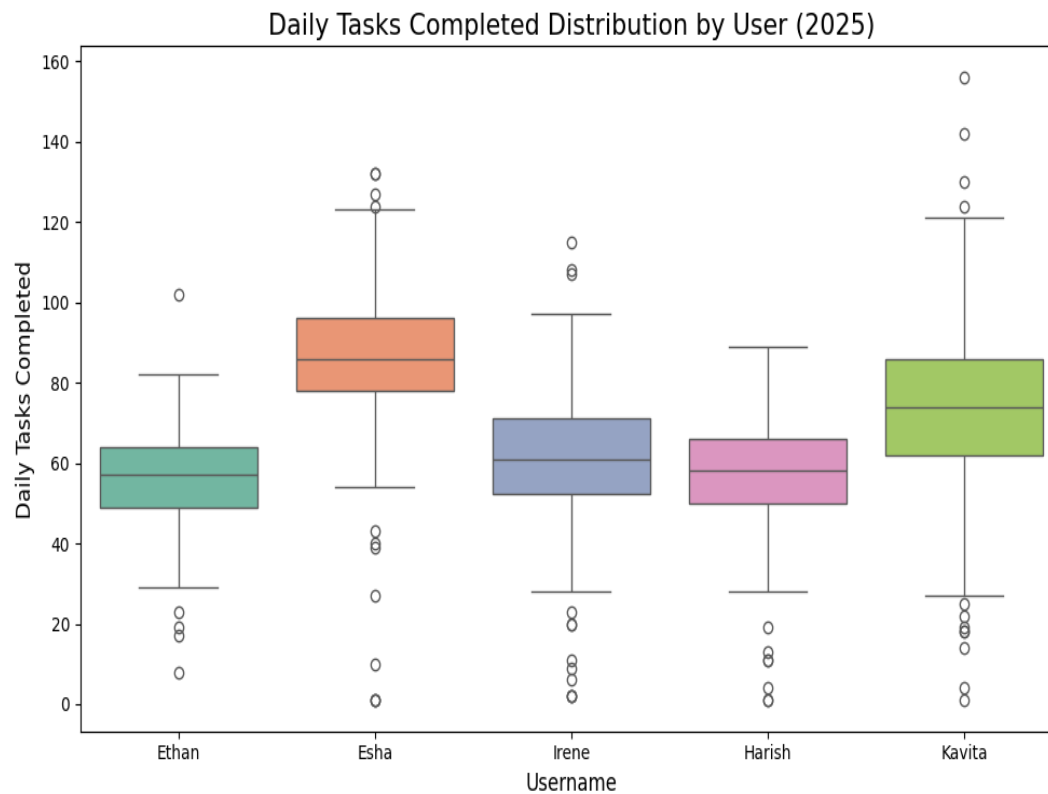
- Esha & Kavita - 70 to 80 % of average time
- Ethan - little over but close to average time
- Irene & Harish - 110 to 115 % of average time



Average Total # of Tasks Completed by Team Members on Daily Basis

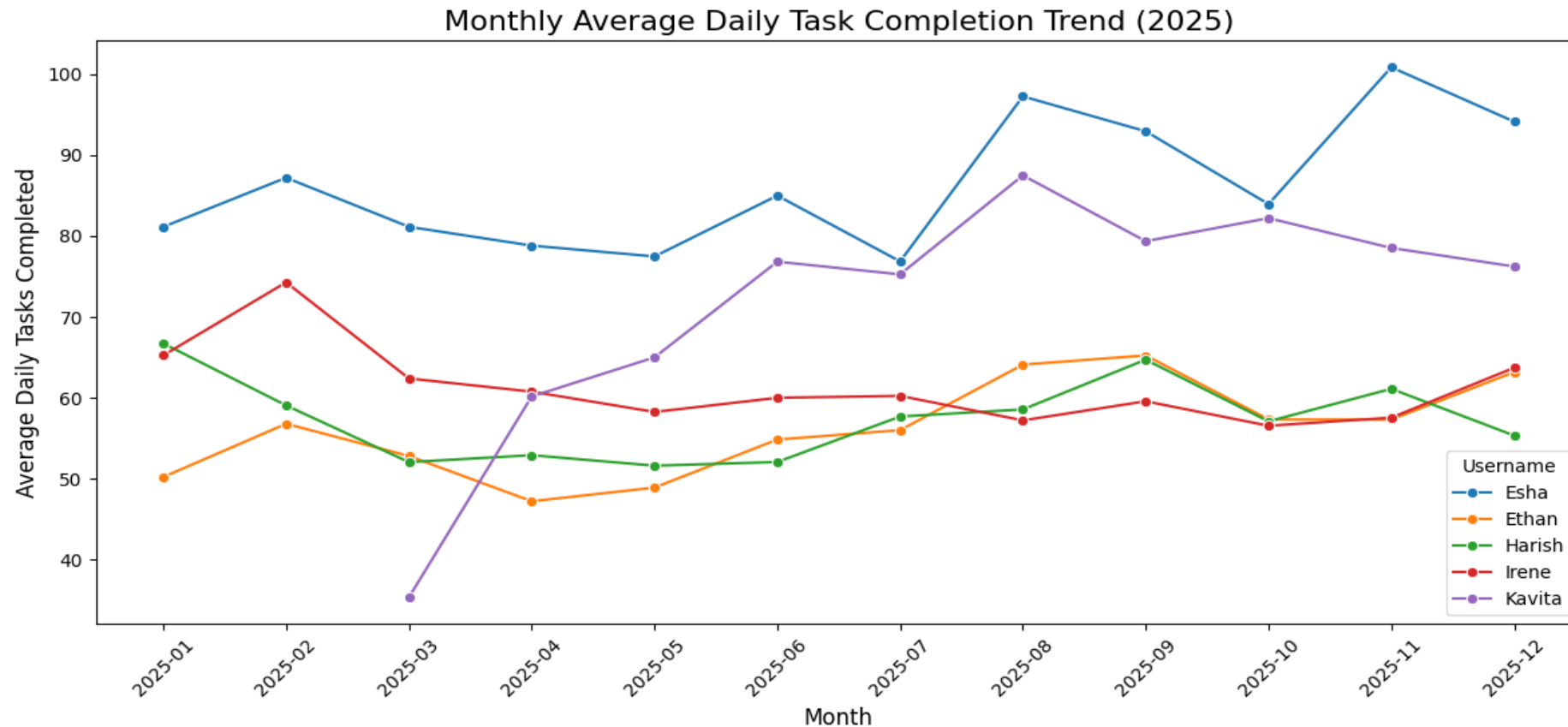
Username	Average_Daily_Tasks
Esha	86
Ethan	56
Harish	57
Irene	61
Kavita	73

Average Daily Tasks: 67



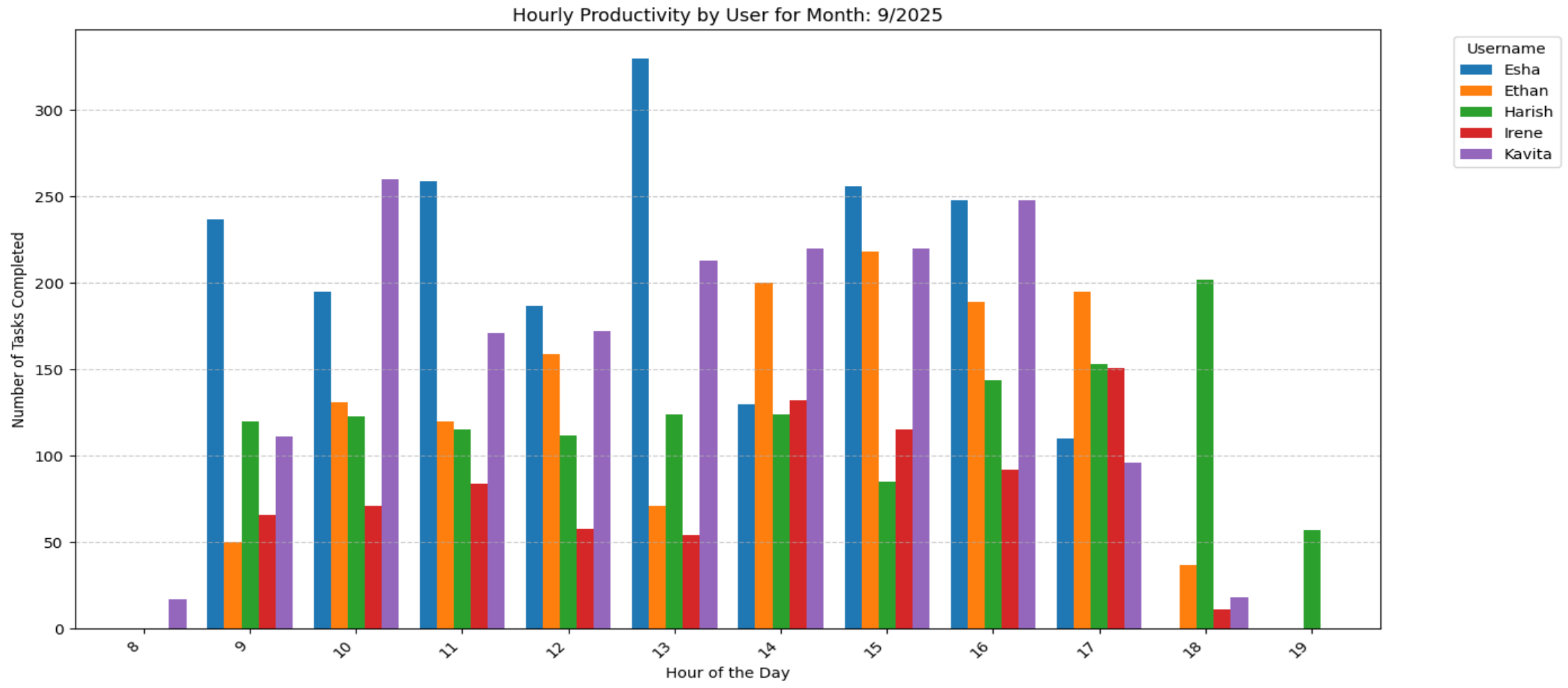
- Esha & Kavita - 10 to 30 % more than number of average daily tasks
- Irene, Harish & Ethan - 10 to 15 % less than number of average daily tasks

Average Total # of Tasks Completed by Team Members on Monthly Basis



When comparing the averages of monthly tasks completed by a team member, Esha has the higher number of tasks month by month followed by Kavita who started in March 2025 while Irene, Harish and Ethan's productivity plateaued and were below Esha and Kavita's.

Hourly Productivity of Team Members for a Random Month



As you see on the graph Hourly productivity was consistent throughout the day for some of the users when compared to others.

Determine factors influencing higher number of tasks not completed

Notification & Clinical Review Task Comparison

For Notification Task

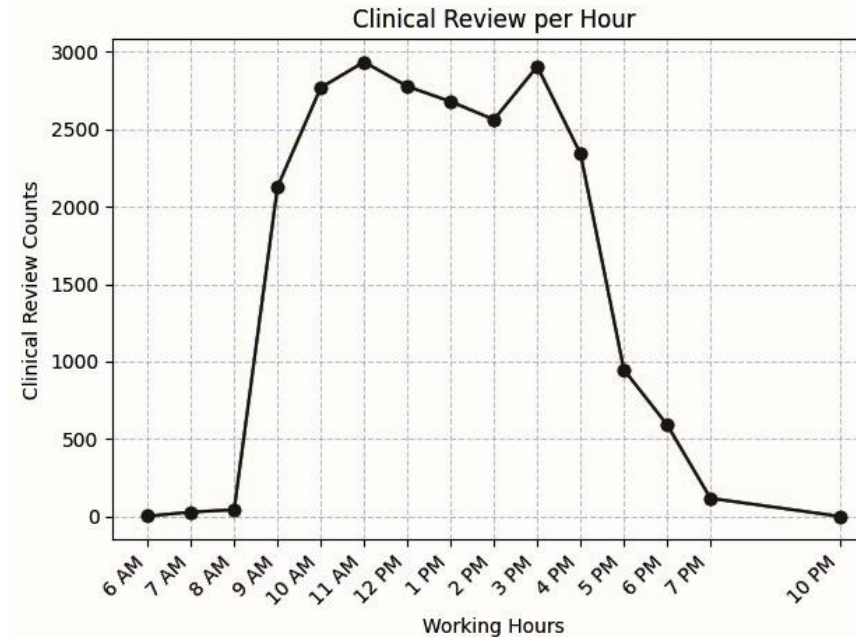
Most productive hour: 17:00 PM

Least productive hour: 7:00 AM

For Clinical Review Task

Most productive hours: 11:00 AM & 15:00 PM

Least productive hour: 22:00 PM



*** Overall, low productive time within shift 9-5:30 was observed between 9 to 10 AM as well as 12 to 2 PM ***

Identify areas of opportunity to increase productivity

To maintain strict CMS 24 to 72-hour turnaround rules, workflows must be structured around CMS Prior Authorization timelines. The primary factors driving uncompleted tasks are misaligned scheduling and after-hours productivity spikes, which cause delays in required clinical reviews and 2nd attempt notifications.

Key Factors Causing Uncompleted Tasks

- **Misaligned Work Patterns:** The data shows that productivity spikes at the end of the day rather than being evenly distributed. Because the 24 to 72-hour CMS clock ticks continuously, waiting until the late afternoon to process cases means there is insufficient runway to complete secondary steps (like final determinations or physician/member follow-ups).
- **Extended-Hour Dip:** While offering a flexible work-life balance is ideal, the data highlights that task resolution drops significantly outside normal business hours. Working outside peak times may interrupt communication with medical offices or members.
- **Clinical Review Bottlenecks:** Because clinical reviews demand rigorous medical necessity evaluations, attempting to tackle them at the end of the shift frequently leads to a backlog of uncompleted reviews and potential compliance risks.

Actionable Strategies to Prevent Compliance Risks

- **Shift Distribution:** Consider implementing a morning-heavy queue for complex clinical reviews. Tasking your most productive hours (start of day) with the highest-risk, time-sensitive clinical items guarantees the team has a buffer to complete subsequent notifications.
- **Synchronous Case Management:** Ensure 2nd attempt notifications are handled at pre-scheduled mid-day intervals. This removes the reliance on after-hours productivity, resulting in faster answers and less friction for both providers and member.
- **Task Segmentation:** Separate "Notifications" from "Clinical Reviews" in the dashboard. Assign dedicated staff to continuous notification-clearing windows throughout the workday to avoid bottlenecks caused by waiting on the clinical review team.

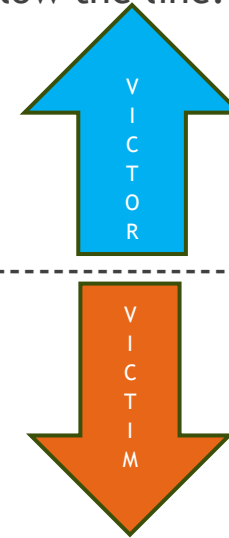
For Team: Ownership, Accountability & Responsibility



Are you above or below the line?

Ownership
Accountable
Responsible

Blame
Excuses
Denial



- ▶ Ownership mindset & behavior
- ▶ Doing the job Vs. achieving results
- ▶ Blaming others Vs. taking accountability
- ▶ Responsibility: a foundation for success

Factors & Strategies for remote leadership



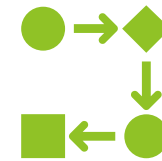
Establish Clear Goals

- Set roadmap
- Provide resources
- Clear communication
- Trust & Empowerment
- Transparency



Regular Check-ins

- Periodic follow ups
- Celebrate milestones
- Address challenges
- Building relationships
- Be adaptive and flexible



WorkiQ

- Tracks time spent in work and non-work applications
- Measures idle time (e.g., breaks, lunch)
- Shows overall activity patterns

Empower remote team members to take ownership of their work

Summary

- ▶ Working remotely isn't as easy as you think
- ▶ Require lots of self control
- ▶ Clear communication
- ▶ Trust your people
- ▶ When in doubt, speak openly
- ▶ Care about the team
- ▶ Support work-life balance

Questions

